# Emily Lei Kang | Curriculum Vitae

Division of Statistics and Data Science, Department of Mathematical Sciences
University of Cincinnati, Cincinnati, Ohio 45221-0025
☎ (513) 556 5138 • ✉ kangel@ucmail.uc.edu

## Education

**The Ohio State University** **Columbus, OH**
*PhD in Statistics* *2004–2009*

**Tianjin University** **Tianjin, China**
*BS in Applied Mathematics* *2000–2004*

**Nankai University** **Tianjin, China**
*BA in Finance* *2001–2004*

## Research Interests

- Fundamental methodological research on statistical models and computational statistics for large data with high dimensionality and complex dependence structure, in particular spatial, spatio-temporal and social network data
- Uncertainty quantification, parameter calibration and experiment design for computer experiments
- Bayesian methodology and modeling with applications in environmental, climate, biological sciences and engineering

## Professional Experience

**Department of Mathematical Sciences** **University of Cincinnati**
*Professor* *2021–present*

**Department of Mathematical Sciences** **University of Cincinnati**
*Associate Professor* *2016–2021*

**Statistical and Applied Mathematical Sciences Institute (SAMSI)** **Research Triangle Park, NC**
*Visiting Research Fellow* *2017 Fall*

**Department of Mathematical Sciences** **University of Cincinnati**
*Assistant Professor* *2011–2016*

## Awards and Honors

**2019**: Early Investigator Award from the American Statistical Association Section on Statistics and the Environment

**2017,2018**: Nominated for the George Barbour Award for Good Faculty-Student Relations, University of Cincinnati

**2013**: Travel Award from the Association of Women in Mathematics (AWM)-National Science Foundation (NSF)

**2012**: Faculty Research Fellowship from the University Research Council (URC) at University of Cincinnati

## *Selected* Publications (* denotes a current/former student co-author.)

1. Ren, S.*, Wang, P. and **Kang, E.L.** (2025) A Practical Tool for Visualizing and Measuring Model Selection Uncertainty, *Stat*, in press.
2. Cheng, S.*, Konomi, B. A., Karagiannis, G., and **Kang, E. L.** (2024) Recursive nearest neighbor co-kriging models for big multi-fidelity spatial data sets. *Environmetrics*, `https://doi.org/10.1002/env.2844`.
3. Konomi, B. A., **Kang, E. L.**, Almomani, A.*, and Hobbs, J. (2023) Bayesian Latent Variable Co- kriging Model in Remote Sensing for Quality Flagged Observations, *Journal of Agricultural, Biological and Environmental Statistics*, 28, 423-441.
4. Kalmus, P., Ekanayaka, A.*, **Kang, E. L.**, Baird, M., and Gierach, M. (2022) Past the precipice? Projected coral habitability under global heating. *Earth's Future*, 10, e2021EF002608, `https://doi.org/10.1029/2021EF002608`.
5. Cheng, S.*, Konomi, B. A., Matthews, J. L., Karagiannis, G., and **Kang, E. L.** (2021) Hierarchical Bayesian nearest neighbor co-kriging Gaussian process models; an application to intersatellite calibration. *Spatial Statistics*, 44, `https://doi.org/10.1016/j.spasta.2021.100516`.
6. **Kang, E. L.**, Li, M.*, Cawse-Nicholson, K., and Braverman, A. (2021) Modeling large multivariate spatial data with a multivariate fused Gaussian process. *Journal of the Indian Statistical Association*, 59 (2), 1-38,

https://www.indstatassoc.org/journal-jisa/previous-volumes/dec-2021-volume-592.

7. Ma, P.*, Mondal, A., Konomi, B. A., Hobbs, J., Song, J. J. and **Kang, E. L.** (2021) Computer model emulation with high-dimensional functional output in large-scale observing system uncertainty experiments, *Technometrics,* 64, 65–79.
8. Cawse-Nicholson, K., Braverman, A., **Kang, E. L.**, Li, M.*, Johnson, M., Halverson, G., Anderson, M., Hain, C., Gunson, M., and Hook, S. (2020) Sensitivity and uncertainty quantification for the ECOSTRESS evapotranspiration algorithm - DisALEXI, *International Journal of Applied Earth Observation and Geoinformation*, https://doi.org/10.1016/j.jag.2020.102088
9. Ma, P.* and **Kang, E. L.** (2020) A fused Gaussian process for very large spatial data, *Journal of Computational and Graphical Statistics*, 29, 479-489. *[Early version wins the Student Paper Competition in the ICSA Applied Statistics Symposium]*
10. Ren, S.*, **Kang, E. L.**, and Lu, J. L. (2020) MCEN: a method of simultaneous variable selection and clustering for high dimensional multinomial regression, *Statistics and Computing*, 30, 291-304.
11. Konomi, B. A., Hanandel, A. A.*, Ma, P. *, and **Kang, E. L.** (2019) Computationally efficient nonstationary nearest neighbors Gaussian processes using data-driven techniques. *Environmetrics*, 30:e2571, https://doi.org/10.1002/env.2571.
12. Li, M.*, and **Kang, E. L.** (2019) Randomized algorithms of maximum likelihood estimation with spatial autoregressive models for large-scale networks. *Statistics and Computing*, 28, 1165-1179. *[Winner of 2019 student paper competition of ASA Business & Economic Statistics Section]*
13. Ma, P.*, **Kang, E. L.**, Braverman, A., and Nguyen, H. (2019) Spatial statistical downscaling for constructing high-resolution nature runs in global observing system simulation experiments. *Technometrics*, 61, 322-340, https://doi.org/10.1080/00401706.2018.1524791.
14. Shi, H.* and **Kang, E. L.** (2017) Spatial data fusion for large non-Gaussian remote-sensing datasets. *Stat*, 6, 390-404.
15. Shi, H.*, **Kang, E. L.**, Konomi, B. A., Vemaganti, K., and Madireddy, S. (2017) Uncertainty quantification using the nearest neighbor Gaussian process. In *New Advances in Statistics and Data Science*, eds D.G. Chen, Z. Jin, G. Li, Y. Li, A. Liu, and Y. Zhao. ICSA Book Series in Statistics. Springer, 89-107, https://doi.org/10.1007/978-3-319-69416-0_6.

## Completed External Funding

- NSF (PI, with G. Lin, Purdue University and B. A. Konomi, University of Cincinnati, as Co-PIs): Collaborative Research: Inference and Uncertainty Quantification for High Dimensional Systems in Remote Sensing: Methods, Computation, and Applications, 2021-2025. UC Subcontract: $159,947.
- NASA (Co-I): Neighborhood-scale extreme humid heat health impacts, 2022-2025. UC Subcontract: $99,486.
- NASA (Co-I): Ecological Projection Analytic Collaborative Framework (EcoPro), 2022-2025. UC subcontract: $235,682.

## *Selected* Service

- Committee member, Advanced Research Computing (ARC) Advisory Council, University of Cincinnati, 2025
- Committee Chair for the 2025 Student Paper Competition, Section on Statistics and the Environment, American Statistical Association, 2024-2025
- Associate Editor, *Statistics and Data Science in Imaging*, 2024-present
- Associate Editor, *SIAM/ASA Journal on Uncertainty Quantification (JUQ)*, 2024-present
- Editorial Board Member, *Spatial Statistics*, 2021-present
- Associate Editor, *Journal of Statistical Computation and Simulation*, 2013-present
- Conference Program Committee Member, SIAM UQ 2024
- Committee member, American Statistical Association Advisory Committee on Climate Change Policy, 2021-2024
- Program Committee member and North America Representative, Caucus for Women in Data Science and Statistics, 2022-2023
- Program Chair, Section on Statistics and the Environment, American Statistical Association, 2022
- Associate Editor, *Computers & Geosciences*, 2019-2023
- Core Team Member, US Climate Variability and Predictability (CLIVAR) Data Science Working Group, 2019-2022
- Associate Editor, *Technometrics*, 2022-2023
- Proposal and panel reviewer for NASA, NIH, and NSF regularly